Indian Institute of Information Technology Vadodara

Unified Multi-Modal Image Synthesis for Missing Modality Imputation

CS-637 Image Processing

Presented By:

Vedangi Solanki (20251603020)
Jeni Navadiya (20251602025)
Eshaan Ravindra Patil (20251603008)

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Introduction

- Multi-modal MRI plays a critical role in medical diagnosis and treatment planning.
- Each modality captures distinct tissue characteristics and pathological information.
- However, complete multi-modal acquisition is often unavailable in clinical practice.
- Missing modalities degrade the performance of automated analysis systems.
- This paper proposes a unified deep learning framework to synthesize missing modalities from arbitrary available inputs.

What is the problem?

- Multi-modal MRI is essential for accurate medical diagnosis.
- In real-world clinical settings, some imaging modalities are often missing.
- Most existing deep learning models cannot handle arbitrary combinations of missing modalities.
- Therefore, a unified framework is required to synthesize missing modalities while preserving structural consistency and pathological details.

Goal of the Paper

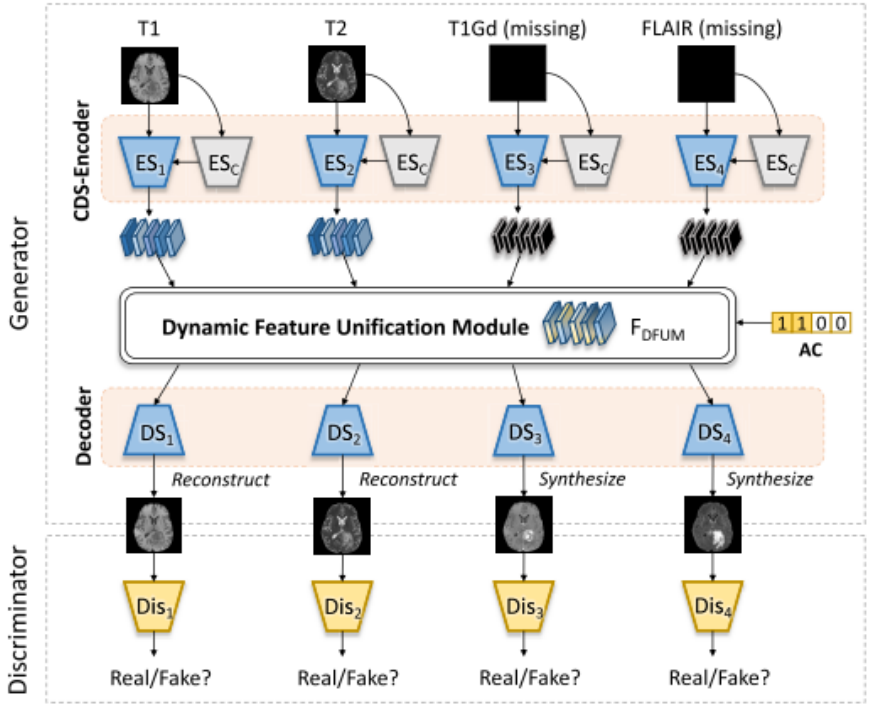
- To develop a unified deep learning framework for missing modality imputation.
- To generate realistic and accurate MRI modalities from arbitrary available inputs.
- To preserve anatomical structures and modality-specific details during synthesis.
- To enhance performance in downstream tasks such as tumor segmentation.

Related Work

- **One-to-One Synthesis:** Generate one modality from a single input using CNNs, GANs, or diffusion models. *Limitation: Requires separate models for each modality pair.*
- **Many-to-One Synthesis:** Fuse multiple input modalities to generate one target modality. *Limitation: Fixed input combinations, limited flexibility.*
- **Unified Synthesis:** Single model handles arbitrary missing-modality scenarios. *Limitation: May lose fine details due to simple feature fusion.*

Proposed Work

- **Unified GAN Framework**
  Synthesizes missing MRI modalities
- **CDS Encoder**
  Extracts shared + specific features
- **DFUM**
  Hard (Max) + Soft (Attention) fusion



Architecture of the Proposed Model

- **Input:** Multi-modal MRI with random missing modalities (Availability Condition vector).
- **Generator (GAN-based):**
  - **CDS-Encoder:** Separate modality-specific streams + shared common stream.
  - **DFUM:** Dynamically integrates features using: Hard fusion (Max) + Soft fusion (Attention).
  - **Multi-Stream Decoder:** Reconstructs available modalities and synthesizes missing ones.
- **Discriminators:** PatchGAN-based discriminators for each modality to ensure realistic outputs.

GAN Framework

- The model follows a **Generative Adversarial Network (GAN)** architecture.
- **Generator (G):** Takes zero-imputed multi-modal inputs and synthesizes missing modalities.
- **Discriminators (D):** Four PatchGAN discriminators (one per modality) distinguish real vs synthetic images.
- The network is trained using a **hybrid loss**:
  - Synthesis Loss (L_{syn})
  - Reconstruction Loss (L_{rec})
  - Adversarial Loss (L_{adv})

CDS-Encoder

- Designed to extract both:
  - **Modality-invariant features** (shared anatomy)
  - **Modality-specific features** (unique tissue details)
- Consists of:
  - Four modality-specific encoding streams: E_{S_i}
  - One shared common encoding stream: E_{SC}
- Features are progressively fused at deeper layers.
- Outputs multi-scale features for each modality.

CDS-Encoder: Feature Representation

- Each input modality X_i is processed by:

$$E_{S_i}(X_i) \quad \text{(modality-specific stream)}$$

$$E_{SC}(X_i) \quad \text{(shared common stream)}$$

- Multi-scale feature maps are extracted:

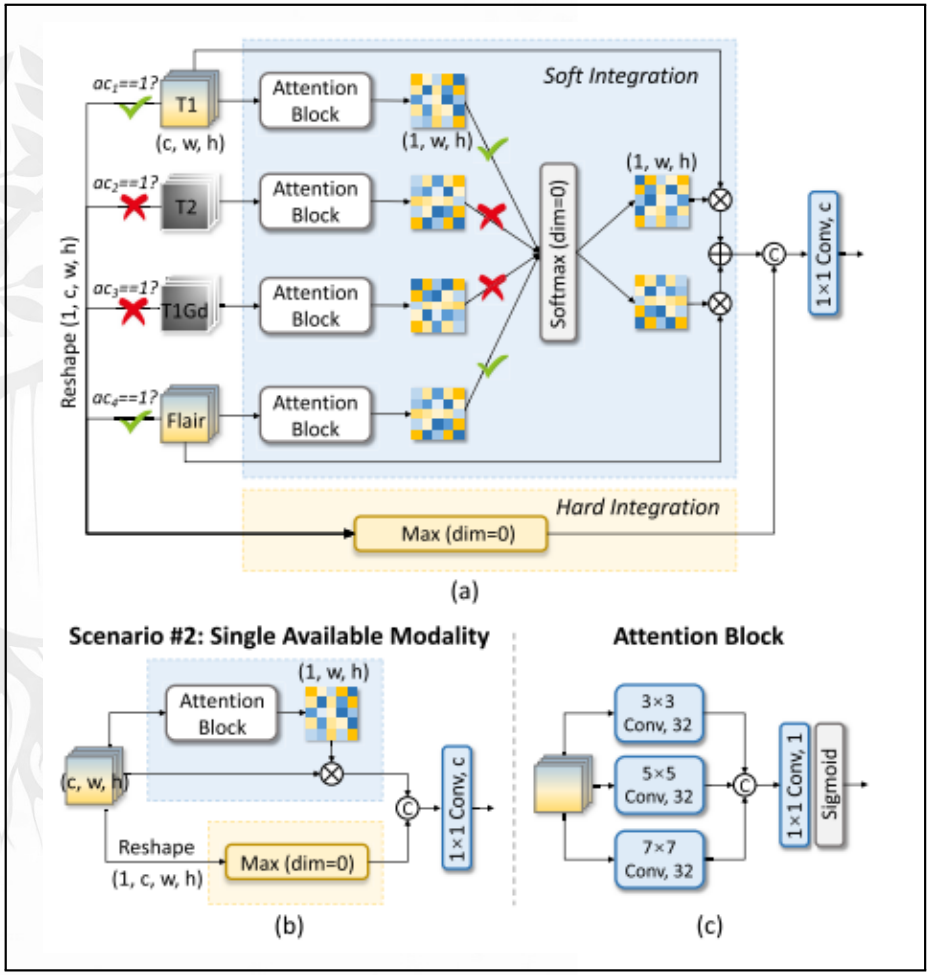
$$\{F_i^{(s)}\}_{i=1}^{4}, \quad s \in \{1, 2, 3, 4, 5\}$$

- At each scale, features are fused using:

$$F_i^{(s)} = Conv_{1\times1}\big(Concat(F_{S_i}^{(s)}, F_C^{(s)})\big)$$

Dynamic Feature Unification Module (DFUM)

- Designed to integrate features from a varying number of available modalities.
- Addresses limitation of simple Max fusion (which may lose subtle structural details).
- Consists of:
  - **Hard Integration** (Max operation)
  - **Soft Integration** (Attention-based fusion)



DFUM architecture showing hard + soft fusion

Dynamic Feature Unification Module

- Integrates multi-scale features from available modalities:

$$\{F_i^{(s)}\}, \quad a_i \in \{0, 1\}$$

- **Hard Integration (Max Fusion):**

$$F_{hard}^{(s)} = Max\left(Concat\{F_i^{(s)} \mid a_i = 1\}\right)$$

- **Soft Integration (Attention Fusion):**

$$W^{(s)} = \sigma\left(Concat\{Attn(F_i^{(s)}) \mid a_i = 1\}\right)$$

$$F_{soft}^{(s)} = \sum_{i=1}^{4} a_i \, W_i^{(s)} \otimes F_i^{(s)}$$

DFUM: Final Unified Feature

- Hard and Soft integration are combined:

$$F_{DFUM}^{(s)} = Conv_{1\times1}\left(Concat\{F_{hard}^{(s)}, F_{soft}^{(s)}\}\right)$$

- Ensures:
  - Discrete feature preservation (Hard)
  - Continuous feature weighting (Soft)

Generator Loss

- **Synthesis Loss (for missing modalities):**

$$L_{syn} = \sum_{i=1}^{4} (1 - ac_i)\, L_1(\hat{Y}_i, Y_i)$$

Ensures accurate generation of missing modalities.

- **Reconstruction Loss (for available modalities):**

$$L_{rec} = \sum_{i=1}^{4} ac_i\, L_1(\hat{Y}_i, Y_i)$$

Maintains consistency for available inputs.

Generator Loss

- **Adversarial Loss (Least Squares GAN):**

$$L_{adv} = \sum_{i=1}^{4} (1 - ac_i)(L_2(\hat{Y}_i) - 1)^2$$

Encourages realistic texture generation.

- **Total Generator Loss:**

$$L_{Gen} = \lambda_1 L_{syn} + \lambda_2 L_{rec} + \lambda_3 L_{adv}$$

Weighted combination of pixel accuracy and realism.

Discriminator Loss

- **Least Squares GAN (LSGAN) Objective:**

$$L_{Dis} = \sum_{i=1}^{4}(1 - ac_i)\left[(L_2(Dis_i(\hat{Y}_i))^2 + (L_2(Dis_i(Y_i) - 1)^2\right]$$

- We use a least square loss to optimize the discriminators.

Tumor Segmentation

- After reconstructing missing MRI modalities, the complete set of inputs is used for tumor segmentation.
- A **U-Net based architecture** is employed for pixel-wise classification.
- The model predicts multi-class tumor regions:
  - Background
  - Edema
  - Tumor Core
  - Enhancing Tumor
- Encoder captures high-level features, while decoder restores spatial resolution using skip connections.
- This ensures accurate localization and boundary refinement of tumor regions.

Segmentation Evaluation

- Performance evaluated using **Dice Score**.
- Dice measures overlap between prediction and ground truth:

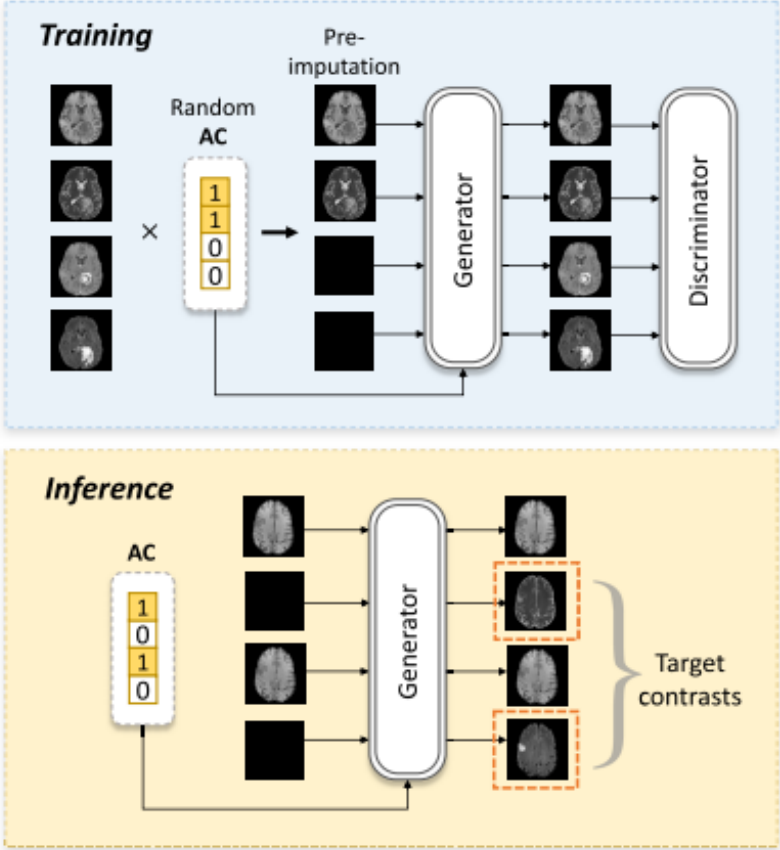
$$Dice = \frac{2|P \cap G|}{|P| + |G|}$$

- Region-wise evaluation (BraTS):
  - WT (Whole Tumor)
  - TC (Tumor Core)
  - ET (Enhancing Tumor)
- Provides accurate assessment of segmentation quality.

Synthesis Training (GAN)

Training Steps

- Load BraTS MRI slices (T1, T2, T1ce, FLAIR)
- Randomly remove modalities (simulate missing data)
- Forward pass:
  - Encoder → DFUM → Generator
- Generate missing modalities
- Compute losses:
  - L_{syn} (missing)
  - L_{rec} (available)
  - L_{adv} (GAN)
- Update Generator + Discriminator



GAN training pipeline

Segmentation Training

Training Steps

- Input: Complete modalities (real + synthesized MRI)
- Ground truth masks available in **seg.nii.gz** format (BraTS dataset)
- Preprocessing:
  - Convert 3D volumes to 2D slices
- Forward pass through **U-Net**
- Predict multi-class segmentation mask:
  - Background
  - Edema
  - Tumor Core
  - Enhancing Tumor
- Compute loss:
  - Cross-Entropy Loss
  - Dice Loss
- Backpropagation and weight update

Datasets Used

- **BraTS 2019 Dataset**
  - Brain tumor MRI dataset
  - Modalities: T1, T2, T1Gd, FLAIR
  - 250 training, 20 validation, 20 testing subjects
  - Middle 80 axial slices (192×192)
- All images are mean-normalized before training.

Evaluation Metrics

- **1. Peak Signal-to-Noise Ratio (PSNR):**

$$PSNR(\hat{y}, y) = 10 \log_{10} \left(\frac{max^2(\hat{y}, y)}{||\Omega|^{-1} |\hat{y} - y||_2^2} \right)$$

Measures pixel-level reconstruction accuracy. Higher is better.

- **2. Structural Similarity Index (SSIM):**

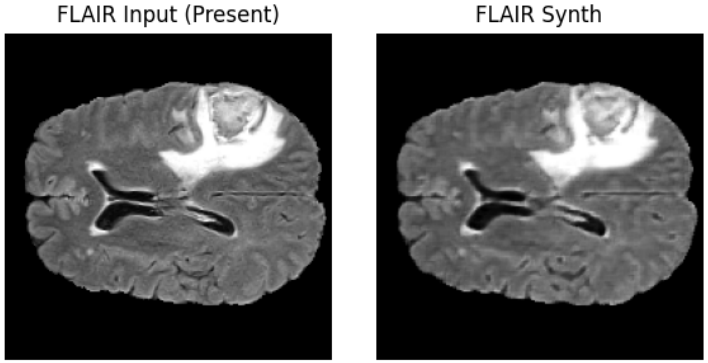
$$SSIM(\hat{y}, y) = \frac{(2\mu_{\hat{y}}\mu_y + c_1)(2\sigma_{\hat{y}y} + c_2)}{(\mu_{\hat{y}}^2 + \mu_y^2 + c_1)(\sigma_{\hat{y}}^2 + \sigma_y^2 + c_2)}$$

Measures structural similarity between generated and real images.

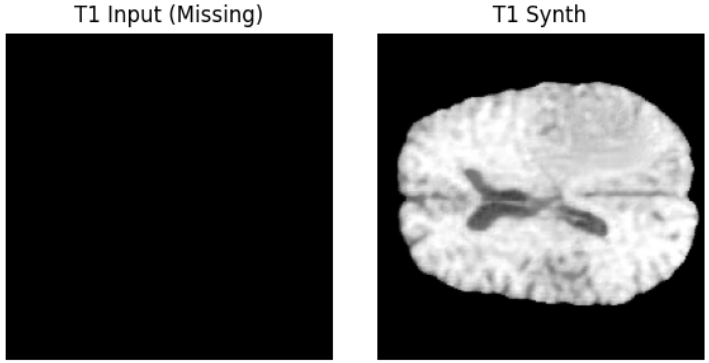
Experimental Results

- **Datasets:** BraTS 2019 (4 modalities)
- **Compared with:** pGAN, Pix2Pix, Hi-Net, MM-Synthesis, MM-GAN, MM-Transformer
- Unified model outperforms one-to-one and many-to-one method achieving highest **PSNR** and **SSIM** in most of them.
- Better preservation of:
  - Tumor morphology
  - Fine anatomical structures
  - Intensity consistency

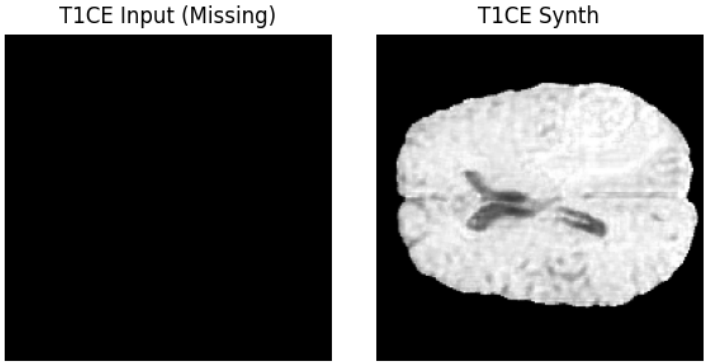
Qualitative Results (Generated Modalities)



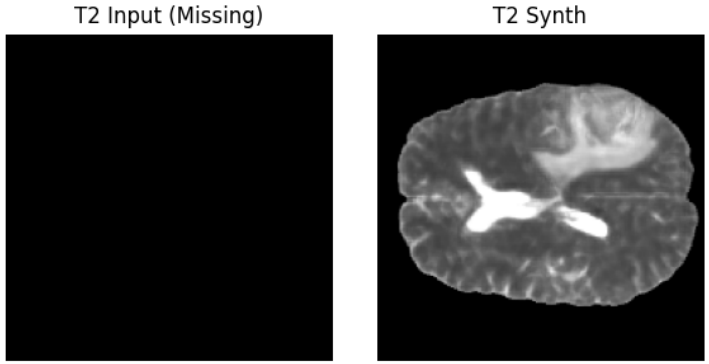
FLAIR modality synthesis



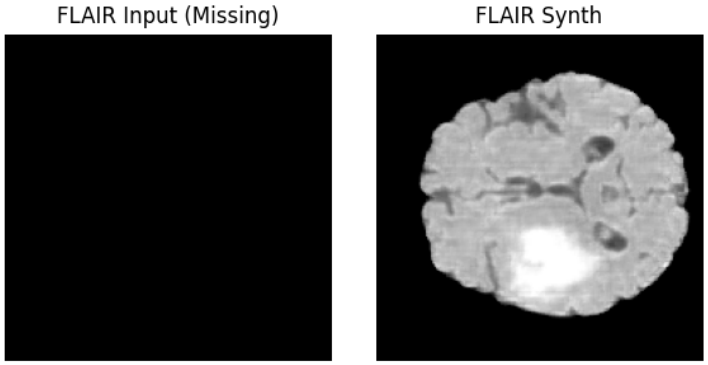
T1 modality synthesis



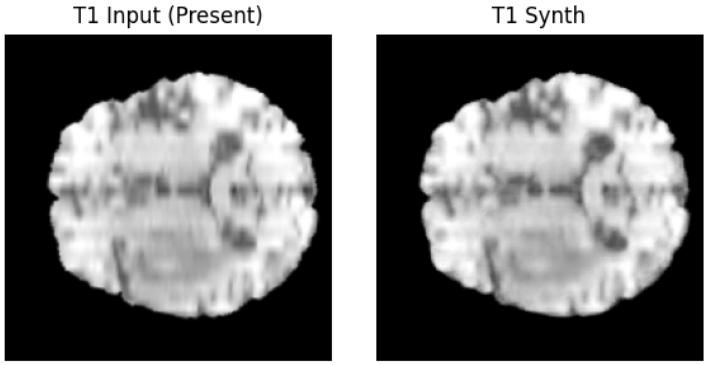
T1CE modality synthesis



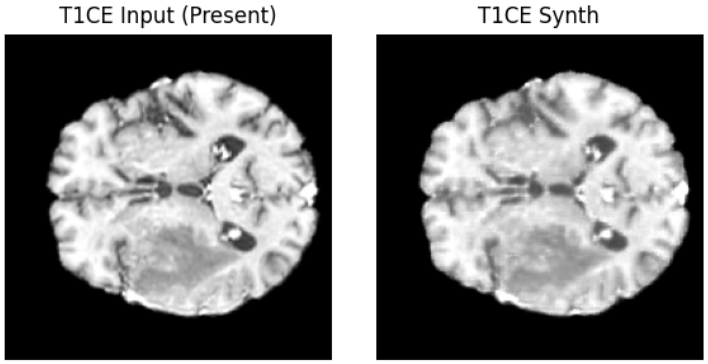
T2 modality synthesis



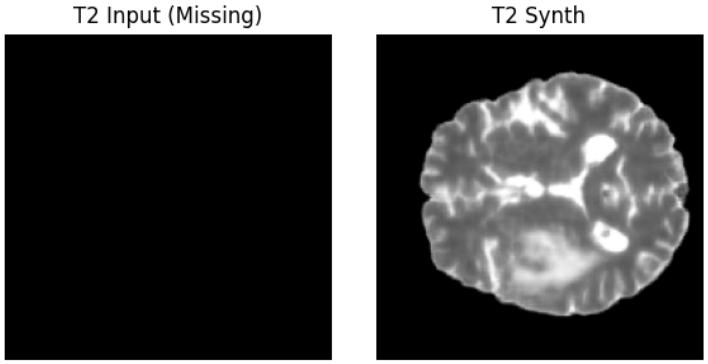
FLAIR modality synthesis



T1 modality synthesis

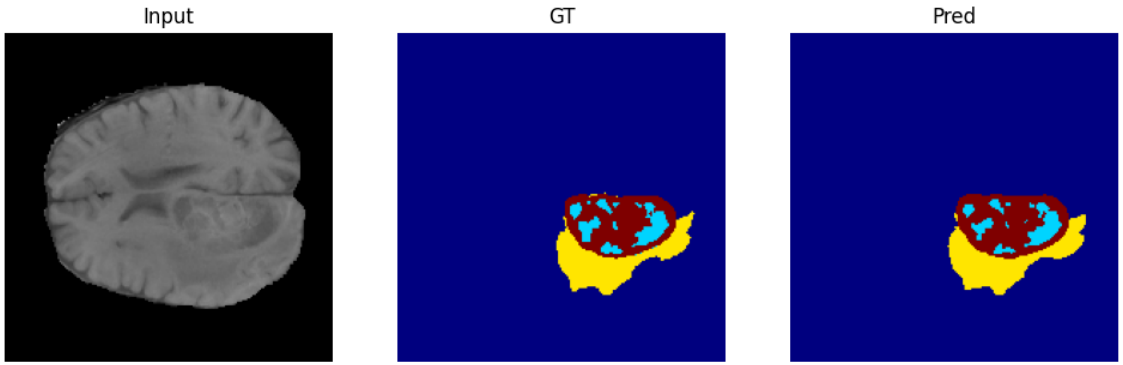


T1CE modality synthesis

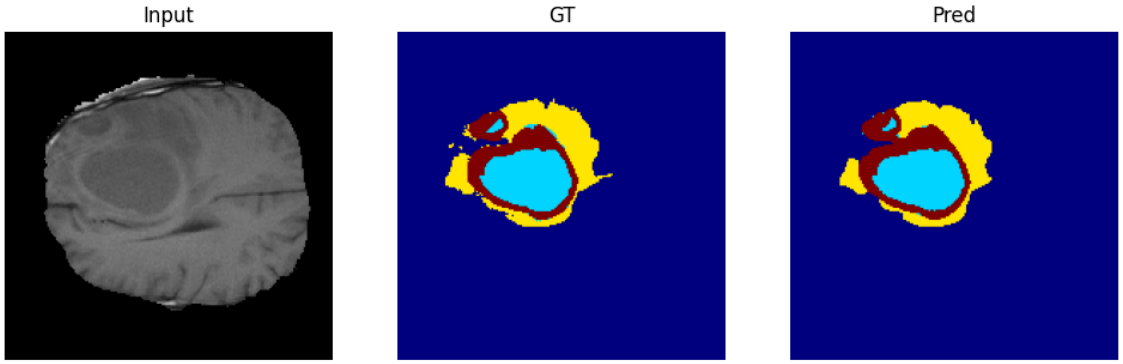


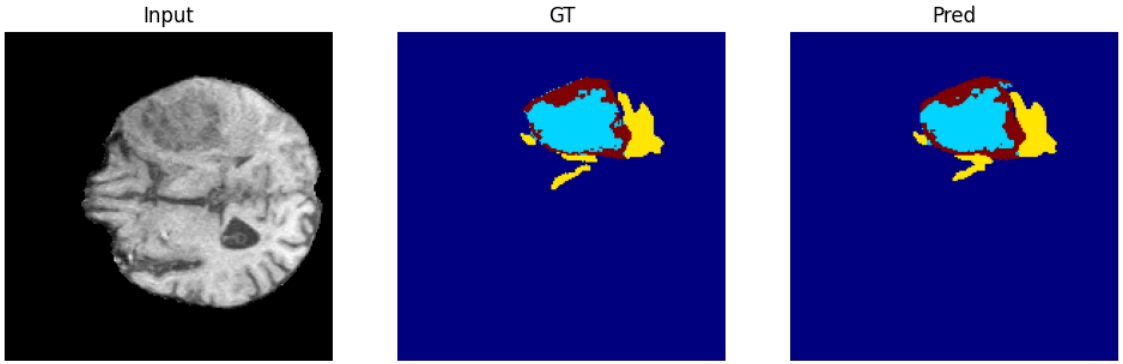
T2 modality synthesis

Segmentation Results



Segmentation Results





Training Metrics

Epoch	PSNR	SSIM
1	16.43	0.2300
2	19.66	0.2521
3	18.90	0.2707
	⋮	
31	27.22	0.8394
32	22.78	0.7786
33	27.54	0.8535
	⋮	
57	24.37	0.8291
58	23.72	0.8207
59	24.27	0.7859

PSNR & SSIM during training

Epoch	WT	TC	ET
1	0.87	0.82	0.79
2	0.84	0.80	0.76
3	0.89	0.85	0.81
4	0.86	0.81	0.78
5	0.88	0.83	0.80

Dice Score for WT (Whole Tumor), TC (Tumor Core), ET (Enhancing Tumor)

Conclusion and Future Work

Conclusion

- Proposed a unified GAN framework for multi-modal MRI synthesis.
- Introduced CDS-Encoder to model shared and modality-specific features and designed DFUM for dynamic feature integration.
- Achieved superior PSNR and SSIM on BraTS datasets.

Future Work

- Develop lightweight architectures for better scalability.
- Improve 3D synthesis consistency across planes.
- Extend framework for cross-dataset generalization.

References

- L. Shen et al., *Multi-domain image completion for random missing input data*, IEEE TMI, 2021.
- A. Chartsias et al., *Multi-input multi-output CNN for multi-modal medical image synthesis*, 2018.
- A. Sharma and G. Hamarneh, *MM-GAN: 3D Multi-modal GAN for brain MRI synthesis*, 2020.
- I. Goodfellow et al., *Generative Adversarial Networks*, NeurIPS, 2014.
- P. Isola et al., *Image-to-Image Translation with Conditional GANs (Pix2Pix)*, CVPR, 2017.

Thank You!

Proposed Work

- **Unified GAN Framework:** A single model synthesizes missing MRI modalities from any combination of available inputs.
- **CDS-Encoder:** Extracts both modality-invariant (shared anatomy) and modality-specific (fine tissue details) features.
- **Dynamic Feature Unification Module (DFUM):** Integrates features from varying numbers of modalities using hard (Max) + soft (attention) fusion.